

DSG1_T3_Progress_Update

Team: MHOB

Project Name: PathEats

Route-Aware Food Discovery Platform

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1. Requirements

Development Methodology: Agile (Scrum)

PathEats spans 5 domains (Consumer, Vendor, Global Admin, Business Assistance, Developer Admin) requiring parallel development. Agile enables:

- **Iterative delivery** — working increments every sprint
- **Parallel feature work** — via feature branches merged into dev
- **API-first contracts** — agree on endpoints before implementation
- **Layered architecture** — Route → Controller → Service → Repository → DB
- **Daily integration** — feature branches merge --no-ff into dev

Functional Requirements & User Stories

ID	Requirement	User Story	Status
FR-01	User registration & login with JWT	As a user, I want to register/log in to access platform features	Done
FR-02	Route-based vendor search (origin → destination)	As a consumer, I want to enter origin/destination to find food vendors along my route	Done
FR-03	Filter vendors by cuisine, price, open status	As a consumer, I want to filter vendors to narrow down choices	Done
FR-04	Vendor detail view (menu, hours, photos, reviews)	As a consumer, I want to view vendor details before deciding	Done
FR-05	Write & read reviews for vendors	As a consumer, I want to leave reviews so others benefit	Done
FR-06	Favorite/bookmark vendors	As a consumer, I want to save favorite vendors	In-memory
FR-07	Vendor stall auto-approved on creation	As a vendor, my stall is immediately visible after creation (no pending approval)	Done
FR-08	Menu item CRUD management	As a vendor, I want to manage my menu items with prices	Done

FR-09	Vendor dashboard with metrics	As a vendor, I want to view stall metrics and performance	Done
FR-10	Vendor location pinpoint on interactive map	As a vendor, I want to set my stall location with a draggable marker	Done
FR-11	Admin user management (list, ban, role change)	As an admin, I want to manage users and enforce platform rules	Done
FR-12	Business Assistance: vendor onboarding config	As a business assistant, I want to configure vendor onboarding (Telegram link, message)	Done
FR-13	Business Assistance: stall CRUD for any vendor	As a business assistant, I want to help vendors create/update/delete their stalls	Done
FR-14	Review moderation (flag, unflag, remove)	As a business assistant, I want to flag/unflag/remove inappropriate reviews	Done
FR-15	Role & permissions management	As an admin, I want to manage platform roles and their privileges	Done
FR-16	Database activity monitor & audit log	As an admin, I want to monitor live database activity and audit history	Done
FR-17	Developer SQL query editor with presets	As a developer admin, I want to run SQL queries against the database	Done
FR-18	System health monitoring & maintenance	As a developer admin, I want to monitor system health and run VACUUM/ANALYZE	Done

Non-Functional Requirements

ID	Requirement	Description
NFR-01	Performance	Route-based vendor scoring responds within 3s; PostGIS spatial indexes on location data
NFR-02	Security	JWT access/refresh tokens, bcrypt password hashing, RBAC middleware per domain
NFR-03	Scalability	Stateless backend, pg.Pool connection pooling, GIST spatial indexes for geo queries
NFR-04	Usability	Responsive design, dark/light theme, interactive MapLibre map with draggable marker
NFR-05	Maintainability	Layered architecture, OOP with dependency injection, separation of concerns
NFR-06	Reliability	PostgreSQL + PostGIS for spatial integrity; audit logging for admin actions
NFR-07	Extensibility	API-first contracts enable independent frontend/backend development

2. Use Case Diagrams

Main Actors

- **Consumer** — End user searching for food along a route
- **Vendor** — Food stall owner managing their business
- **Global Admin** — Platform administrator managing users, roles, and system
- **Business Assistance** — Helps vendors with onboarding, stall CRUD, and review moderation
- **Developer Admin** — Technical admin monitoring system health and database

Use Case Diagram

Actors and their use cases:

- **Consumer:** Register/Login, Search Vendors by Route, Filter & Sort, View Vendor Details & Reviews, Write Reviews, Bookmark Favorites
- **Vendor:** Register/Login, Manage Stalls (CRUD, auto-approved), Manage Menu Items, View Vendor Dashboard, Pinpoint Location on Map
- **Global Admin:** Register/Login, Manage Users, Manage Roles & Permissions, Monitor DB Activity & Audit
- **Business Assistance:** Register/Login, CRUD Stalls for Any Vendor, Configure Onboarding, Flag/Unflag/Remove Reviews
- **Developer Admin:** Register/Login, Run SQL Queries, System Health & Maintenance

Full diagram in here:

https://www.figma.com/board/KzteLkVhbsWYyOTZWvP07R/ERD_PATHEAT?node-id=0-1&t=BAw6zgvtrEXpZ74U-1

3. Activity Diagrams

Workflow 1: Consumer Searches for Food Along a Route

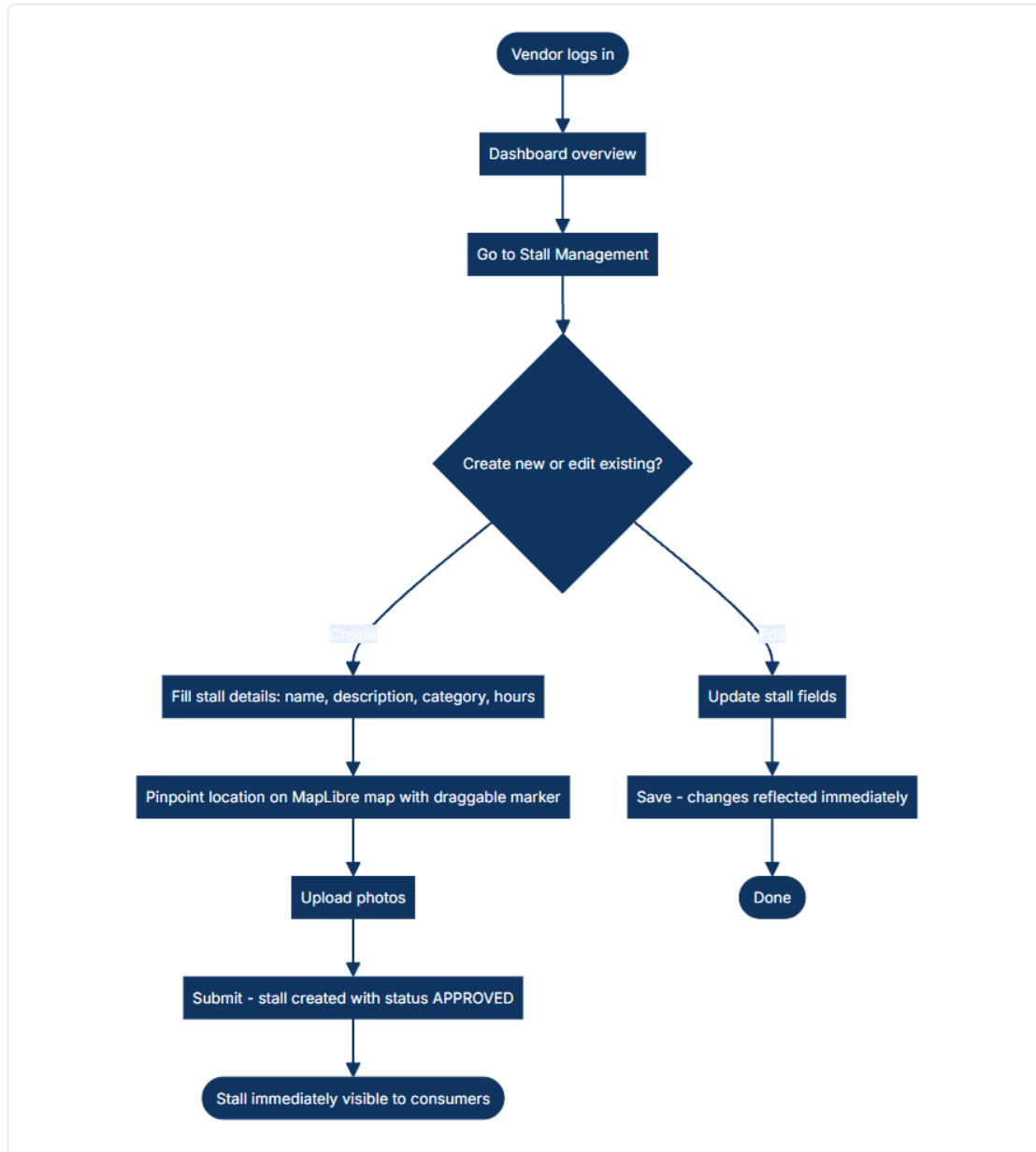
1. Consumer opens the app and enters origin & destination
2. System calculates route via OSRM routing service
3. System queries vendors near the route using PostGIS spatial query
4. Vendors are scored by distance from route, price, rating, and wait time
5. Ranked results are displayed on the map and sidebar
6. Consumer can filter by cuisine, price, open-now, or search by name/menu
7. Consumer selects a vendor → slide-in panel shows details, menu, hours, reviews
8. Consumer can write a review (rating + text) which is saved to the database

No diagram yet.

Workflow 2: Vendor Creates a Stall (Auto-Approved)

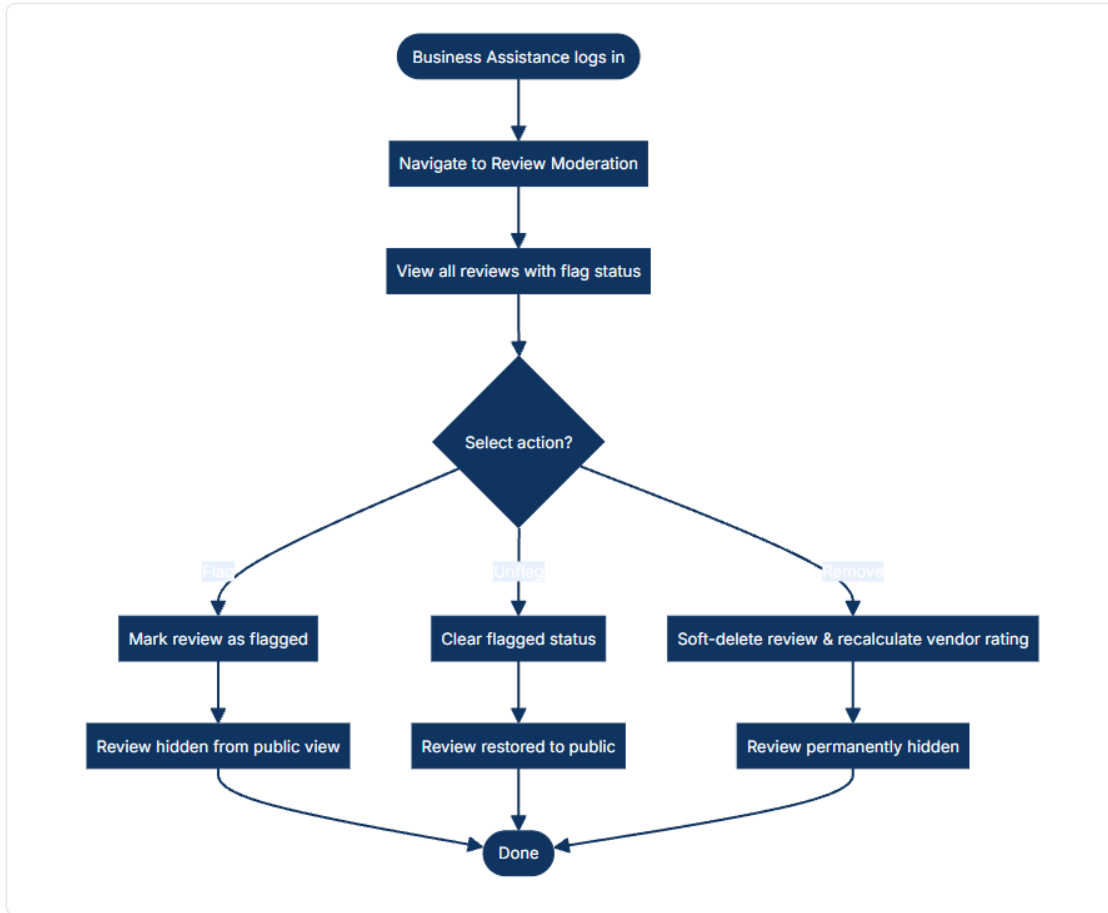
1. Vendor logs in and views dashboard with stall metrics
2. Vendor navigates to Stall Management
3. For a new stall: fills in name, description, category, hours

4. Pinpoints location on an interactive MapLibre map with draggable marker
5. System creates the stall with status **APPROVED** — no pending/approval workflow
6. Stall is immediately visible to consumers searching along routes
7. For existing stalls: vendor can edit details and save changes immediately



Workflow 3: Business Assistance Flags a Review

1. Business Assistance logs in and navigates to Review Moderation
2. All reviews are displayed with their flag status
3. Agent can **flag** a review → marks it as flagged, hidden from public
4. Agent can **unflag** a review → restores it to public view
5. Agent can **remove** a review → soft-deletes it and recalculates vendor rating



4. Team Responsibilities per User Stories

Kong LeakSmey — Consumer Domain, and also help in other admin roles.

- Route-based vendor search UI (origin/dest input, map, results strip)
- Vendor filtering by cuisine, price, open-now, search
- Vendor detail slide-in panel with menu, hours, reviews
- Review writing & display (consumer-facing)
- Favorites & search history UI
- User auth modal (login/signup)
- Map integration (MapLibre GL JS, markers, route polyline)

Leng Layhok — Vendor & Developer Admin

- Vendor stall CRUD pages & API (auto-approved on creation)
- Menu item management (frontend + backend)
- Vendor dashboard with metrics
- Location pinpoint with draggable map marker
- Vendor onboarding & settings pages
- Developer admin: SQL query editor, system health, DB maintenance
- Developer: backup/recovery management

Keo Seavpav — Global Admin & Business Assistance

- Admin user management (list, create, ban, role change)
- Role & permissions management
- Admin stall management (CRUD for any vendor's stalls)
- Review moderation: flag, unflag, soft-delete with rating recalculation
- Vendor onboarding configuration (Telegram link, message)
- Admin audit: live database activity monitor
- Auth system (JWT middleware, RBAC), database schema & PostGIS setup

5. Development Planning

Sprint 1 — Foundation Done

Week 1-2

- Project scaffolding: React + Vite + TypeScript frontend, Express + ESM backend
- Database schema: 11 tables (users, places, reviews, menu_items, etc.) with PostGIS GIST indexes
- Authentication system: register, login, JWT tokens, OTP password reset, RBAC middleware
- Core UI structure: routing, auth context, theme context, shared components
- Backend OOP architecture: repositories, services, controllers with DI

Sprint 2 — Core Features In Progress

Week 3-4

- Consumer route-based vendor search (OSRM routing + PostGIS proximity scoring)
- Interactive MapLibre map: vendor markers, route polyline, draggable marker
- Vendor stall CRUD — auto-approved on creation, no pending workflow
- Vendor dashboard with real metrics (total/open stalls, avg rating)
- Menu item CRUD (global + per-stall scoped)
- Vendor filtering: cuisine, price range, open-now, name/menu search
- Review system: consumer writes reviews, vendor views their reviews

Sprint 3 — Admin & Business Assistance Upcoming

Week 5-6

- Global admin: user management (CRUD, ban, role change)
- Business Assistance: stall CRUD for any vendor (admin-managed)
- Review moderation: flag, unflag, soft-delete with rating recalculation
- Vendor onboarding configuration (Telegram link, message via admin panel)
- Role & permissions management with JSONB privileges
- Admin audit: live pg_stat_activity + audit_log viewer
- Developer tools: SQL query editor with 11 presets, VACUUM/ANALYZE maintenance
- Developer admin: system health dashboard, backup/recovery tracking

Sprint 4 — Polish & Deployment Upcoming

Week 7

- Persist favorites & search history to backend (currently in-memory/localStorage)
- Performance optimization: map marker pooling, query caching, lazy loading
- Error boundary coverage & edge case handling
- Swagger API documentation completion
- Database indexing audit & query optimization
- Deployment to staging/production